

The Macroevolutionary Level

To understand humans:

- compare biology + behavior to other organisms
- see how we fit in biological history
- find our place in the animal kingdom
- what has happened to us over time

Biological Classification - Taxonomy

↳ hierarchical system based on similarity

- make statements about evolutionary or genetic relationships
- follow taxonomical rules

Homologous Structures: similar structures across species
 • caused by common ancestor; **Homologies**

↓
 controversial -
 gives unclear answers:
 anatomical, molecular info
 often conflict

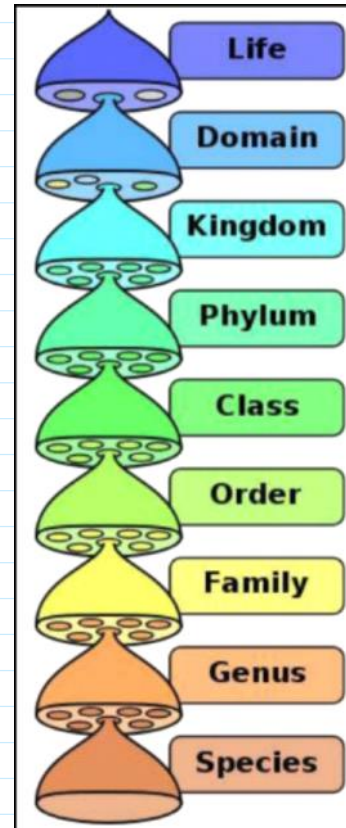
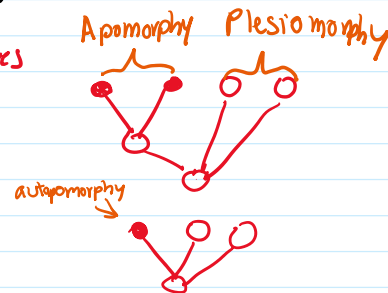
Analogous Structures: similar structures due to function, but not ancestry (superficial, independent)
 • homoplasy: evolutionary development of analogous structures

Evolutionary Systematics: study of homologous traits
 - view broad relationships: **plesiomorphies** (due to ancestral)

Cladistics: emphasis on derived traits (**apomorphies**)

Used to figure out
 common ancestors

- **synapomorphies** - shared, derived traits
- **autapomorphies** - unique, derived traits



Phylogenetic Tree

- links species by common ancestors
- hard to construct if missing nodes

Cladograms

- links species by common traits
- helpful for discovering relationships b/tw similar species

Biological Species Concept ("lumpers")

- As long as animals can breed and produce viable offspring, they should be considered the same species
- not based on physical appearance
- difficult to use on extinct species/fossils → **paleospecies**

Phylogenetic Species Concept ("splitters")

- Species should be defined by the smallest set of organisms that share an ancestor and can be distinguished from other sets
- interbreeding capability is irrelevant

→ Species are often blurred lines and is a human construct
 Species do not always conform to the past/future

Taxonomy of Humans